1.02 Surds

Chapter 1 · Original slide 22

5: Surds

22

Recap:

A surd is a root of a number that does not simplify to a rational number.

Laws:

$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$

$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$

Note: A *rational* number is any which can be expressed as $\frac{a}{b}$ where a, b are integers. $\frac{2}{3}$ and $\frac{4}{1} = 4$ are rational numbers, but π and $\sqrt{2}$ are not.

$\sqrt{3} \times 2$

$3\sqrt{5} \times 2\sqrt{5}$

$\sqrt{8} = \sqrt{4}\sqrt{2}$

$\sqrt{12} + \sqrt{27}$

$\left(\sqrt{8} + 1\right)\left(\sqrt{2} - 3\right)$

Chapter 1 · Original slide 23

Exercise 1E – Pure Y1 Page 13

23

3 Simplify $\sqrt{75} - \sqrt{12}$ giving your answer in the form $a\sqrt{3}$, where a is an integer.

Extension

1 [SMC 2014 Q24] Which of the following is smallest?
$10 - 3\sqrt{11}$; $8 - 3\sqrt{7}$; $5 - 2\sqrt{6}$; $9 - 4\sqrt{5}$; $7 - 4\sqrt{3}$.

2 [SMC 2012 Q21] Which of the following numbers does not have a square root in the form $x + y\sqrt{2}$, where x and y are positive integers?
$17 + 12\sqrt{2}$; $22 + 12\sqrt{2}$; $38 + 12\sqrt{2}$; $54 + 12\sqrt{2}$; $73 + 12\sqrt{2}$.

1.02 Surds

Chapter 1 · Original slide 24

6: Rationalising The Denominator

24

Here's a surd. What could we multiply it by such that it's no longer an irrational number?

$$\sqrt{5}\times$$

Chapter 1 · Original slide 25

Examples

25

$$\frac{3}{\sqrt{2}} =$$

$$\frac{6}{\sqrt{3}} =$$

$$\frac{7}{\sqrt{7}} =$$

$$\frac{15}{\sqrt{5}} + \sqrt{5} =$$

1.02 Surds

Chapter 1 · Original slide 26

More Complex Denominators

26

You've seen 'rationalising a denominator', the idea being that we don't like to divide things by an irrational number.

But what do we multiply the top and bottom by if we have a more complicated denominator?

$$\frac{1}{\sqrt{2}+1} \times$$

Chapter 1 · Original slide 27

More Examples

27

$$\frac{3}{\sqrt{6}-2} \times$$

$$\frac{4}{\sqrt{3}+1} \times$$

$$\frac{3\sqrt{2}+4}{5\sqrt{2}-7} \times$$

1.02 Surds

Chapter 1 · Original slide 28

Test Your Understanding

28

Rationalise the denominator and simplify $\frac{4}{\sqrt{5}-2}$

Rationalise the denominator and simplify $\frac{2\sqrt{3}-1}{3\sqrt{3}+1}$

AQA IGCSE FM June 2013 Paper 1

Solve $y\left(\sqrt{3}-1\right)=8$

Give your answer in the form $a+b\sqrt{3}$ where a and b are integers.

Chapter 1 · Original slide 29

Writing surd expressions in power form

29

This is not in the textbook, but a common type of question in exams, often asked in conjunction with differentiation (a later chapter).

Express $\frac{\left(2+\sqrt{x}\right)^2}{x}$ as powers of x.

Expand.

Write any roots as powers of x.

Split fraction
(some may wish to do this step mentally)

Test Your Understanding

Express $\frac{\left(x+\sqrt{x}\right)^2}{2\sqrt{x}}$ as powers of x.

1.02 Surds

Chapter 1 · Original slide 30

Exercise 1F (Page 15)

30

or alternatively: (not in textbook)

1 Rationalise the denominator and simplify the following:

$$\frac{1}{\sqrt{5}+2} =$$

a

$$\frac{\sqrt{3}}{\sqrt{3}-1} =$$

b

$$\frac{\sqrt{3}+1}{\sqrt{5}-2} =$$

c

$$\frac{2-1}{3\sqrt{3}+4} =$$

$$\frac{5\sqrt{5}-2}{2-3} =$$

d

e

2 Expand and simplify:

$$\left(\sqrt{5}+3\right)\left(\sqrt{5}-2\right)\left(\sqrt{5}+1\right) =$$

3 Rationalise the denominator, giving your answer in the form a+b√3.

$$\frac{3\sqrt{3}+7}{3\sqrt{3}-5} =$$

4 Solve x(4−√6)=10 giving your answer in the form a+b√6.

Solve y(1+√2)−√2=3

Simplify:

$$\frac{\sqrt{a+1}-\sqrt{a}}{\sqrt{a+1}+\sqrt{a}} =$$

5

6